AI-POWERED UPI SCAM & FRAUD DETECTION

Behavioral Anomaly Detection & Graph Analytics Platform

Course: PBL-III (AIM3170)
Category: Application-Based

Domain: FinTech Cybersecurity & ML
Core Model: Isolation Forest

Institution: Manipal University Jaipur
Department: Dept. of AIML

1. Problem Statement & Industry Motivation

Why traditional payment fraud rules fail in real-time UPI ecosystems

The Growing UPI Scam Landscape

- **Massive Volume & Instant Settlement:** UPI processes billions of transactions monthly; irreversible transfers make immediate detection critical.
- **Social Engineering Scams:** Victims are tricked via fake refund links, remote access tools, or phishing QR codes into authorizing payments.
- **Account Takeovers:** Sudden nocturnal balance drainage from unrecognised devices to newly created beneficiary accounts.
- **Mule Account Networks:** Illicit funds are split and rapidly hopped across multiple intermediary accounts to evade freezing.

Limitations of Existing Systems

- **Static Rules & High False Positives:** Simple limits (e.g. amount > $50k) block genuine users while missing sophisticated multi-transfer frauds.
- **Siloed Per-Transaction View:** Traditional systems evaluate single transactions in isolation, blind to temporal bursts and money-flow chains.
- **Lack of Explainability:** Black-box fraud scores lack interpretable behavioral context required for fast investigation triage.
- **Our Core Solution:** An unsupervised batch detection engine combining 30 behavioral signals with graph topology to score and prioritize suspicious activity.

2. Previous Semester (PBL-II) Recap & Transition

Building upon proven competencies in machine learning, user interfaces, and persistence

PBL-II Project Summary (AI Health Assistant)

- **Supervised Disease Prediction:** Multi-label ML classification mapping user symptom vectors to potential diagnoses.
- **Fuzzy Dietary Recommendation:** Automated dietary guidance using token-sort fuzzy matching on medical dietary knowledge bases.
- **Interactive Dashboard & Dual Storage:** Multi-tab Streamlit dashboard with MongoDB persistence and local JSON fallback.
- **Key Competency Acquired:** End-to-end model serialization, input validation, and interactive web dashboard development.

Evolution to PBL-III (UPI Scam Detection)

- **From Supervised to Unsupervised:** Shifting from static symptom classification to unsupervised financial anomaly detection (Isolation Forest) to catch zero-day scam patterns.
- **From Single Input to Batch Analytics:** Ingesting full monthly transaction datasets rather than single form inputs.
- **Leak-Free Feature Engineering:** Developing strict chronological historical profiling to calculate 30 features without future data leakage.
- **Relational Graph Analysis:** Introducing NetworkX graph topology to track multi-hop money flow chains and mule account syndicates.

3. End-to-End System Architecture

Reproducible, data-leak-free batch processing and inference pipeline

1. Raw Batch Ingestion
- 14 Raw Columns
- Full Monthly Datasets
- Automated Format Check

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2. Data Quality & Validation
- Nulls & Duplicate IDs
- Non-negative Values
- Datetime Parsing

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3. Feature Engineering
- Chronological Sorting
- 30 ML Features
- No Future Leakage

4. Preprocessing
- np.log1p (11 log features)
- StandardScaler transform()
- Never refitted

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5. Isolation Forest ML
- Decision Score Output
- Predict: Normal vs Anomaly
- Zero Retraining

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6. Streamlit Dashboard
- Overview KPIs & Filters
- 30-Feature Deep-Dive
- Scored CSV Export

Key Pipeline Design Principle: The pre-trained Isolation Forest and StandardScaler are completely decoupled and treated as immutable artifacts. Preprocessing strictly reproduces the Jupyter training configuration without modifying the models.

4. Feature Engineering: The 30 Model Features

Vectorized extraction strictly prior to current transaction to avoid future leakage

Category	Features Included	Computation Logic & Prevention of Leakage
Direct & Temporal	amount, transaction_step, sender_old_balance, sender_new_balance, receiver_old_balance, receiver_new_balance, device_change_flag, new_device_flag, transaction_hour, is_night	Extracted directly from CSV or parsed from transaction timestamp (night: 22:00–06:00).
Historical Interaction Counts	sender_txn_count_before, same_step_txn_count_before, sender_receiver_txn_count_before, sender_unique_recipients_before, receiver_txn_count_before, receiver_unique_senders_before	Vectorized groupby.cumcount() and cumulative unique recipient sets strictly counting events *prior* to current transaction.
Expanding Statistics & Ratios	sender_balance_ratio, time_since_previous_sender_txn, sender_amount_mean_before, sender_amount_std_before, amount_deviation, amount_zscore	Expanding mean, variance, and Z-score calculated over previous transactions only. Sentinel -1 for first transaction.
Rolling Velocity & Chains	txns_last_1h, txns_last_6h, txns_last_24h, rapid_transaction_flag, hop_number, chain_length, chain_duration_steps, transaction_frequency	O(N log N) rolling count using binary search (np.searchsorted) over epoch timestamps. Multi-hop money flow tracking.

5. Interactive Batch Dashboard (Streamlit)

Operational investigation interface organized into 6 functional sections

Section 1: Upload & Validation
- Drag-and-drop CSV uploader
- Automated schema checks (14 cols)
- 1-click Demo Dataset loader

Section 2: Dataset Overview
- Total Volume & Anomaly Rate
- Min/Max Anomaly Scores
- Highest-Risk Transaction ID

Section 3: Suspicious Table
- Sorted by Anomaly Score ascending
- Filters: Risk Level, Prediction, Type
- Limits: Top 10, 25, 50, 100, All

Section 4: Visual Analytics
- Normal vs. Anomaly breakdown
- Continuous score histogram
- Step trends & Transaction types

Section 5: Transaction Inspector
- Single transaction deep-dive
- Full 30-feature vector viewer
- Raw vs. Transformed values

Section 6: Scored Exports
- Complete Scored Dataset CSV
- Anomalies-Only Filtered CSV
- Audit-ready data structures

Risk Level Tiers: **High Risk** (Score < -0.05) | **Medium Risk** (-0.05 ≤ Score < 0.0) | **Normal / Low Risk** (Score ≥ 0.0)

6. Automated Verification & Technical Integrity

Rigorous test suite verifying zero model drift, exact feature sequence, and export stability

15/15 Verification Suite Results (test_batch_app.py)

✓ **Test 1 & 2:** 500-transaction benchmark loaded and all 14 required columns validated.

✓ **Test 3, 4 & 5:** Exactly 30 features generated matching feature_config.pkl in sequence.

✓ **Test 6:** Exact 11 log features transformed using np.log1p().

✓ **Test 7 & 8:** StandardScaler.transform() executed; scaler was **NOT refitted**.

✓ **Test 9 & 10:** Isolation Forest loaded; model was **NOT retrained**.

✓ **Test 11 & 12:** 100% of transactions scored and assigned discrete predictions.

✓ **Test 13, 14 & 15:** Original columns preserved; complete and filtered CSVs exported.

Sample Test Scenarios & Model Outputs

- **Normal Daytime Payment:** Standard volume, matching primary device, low balance ratio
  ➔ **Score:** +0.0237 | **Prediction:** +1 (Normal / Low Risk)
- **Suspicious Nocturnal Takeover:** Unrecognized new device at 03:00 AM draining 100% of balance to new recipient
  ➔ **Score:** -0.1428 | **Prediction:** -1 (High Risk)
- **Rapid Multi-Hop Layering:** Large transfer received and forwarded downstream within 6 minutes
  ➔ **Score:** -0.0969 | **Prediction:** -1 (High Risk)

7. Ongoing Work & Post-Midterm Roadmap

Advancing from tabular behavioral anomaly detection to relational graph analysis

Milestone	Target Deliverable	Technical Focus	Status
Milestone 1	NetworkX Graph Layer	Account node creation, degree centralities, and mule path cycle detection.	**In Progress**
Milestone 2	Combined Multi-Signal Risk Engine	Synthesizing behavioral anomaly scores with graph topology metrics into a unified 0-100 score.	**Upcoming**
Milestone 3	Interactive Network Visualizer	Interactive node-link visualizer rendering suspicious money flow paths and hubs.	**Upcoming**
Milestone 4	MongoDB Database Persistence	Storing uploaded datasets, investigation audit history, and flagged account registries.	**Upcoming**
Milestone 5	Benchmark Evaluation & Report	Comprehensive precision/recall analysis, false-positive benchmarking, and final documentation.	**Upcoming**

8. Project Summary & Key Takeaways

Recap of progress achieved and open discussion points with mentor

Key Achievements Till Mid-Term

- **Production-Ready Batch Engine:** Built high-performance, leak-free feature engineering calculating 30 ML features from raw monthly data.
- **Model Integrity:** Successfully integrated pre-trained Isolation Forest and StandardScaler without drift or refitting.
- **Rich Streamlit Dashboard:** Delivered interactive triage UI with KPI metrics, filters, charts, deep-dive inspectors, and exports.
- **100% Test Validation:** Verified end-to-end reliability across 15 automated validation checks.

Discussion Points with Mentor

- **Graph Scalability:** Tuning NetworkX graph construction parameters for large-scale multi-thousand transaction networks.
- **Risk Weighting Formula:** Defining optimal weighting between behavioral Isolation Forest scores (60%) and graph centrality metrics (40%).
- **Evaluation Metrics:** Finalizing benchmark evaluation strategy on synthetic fraud scenarios.
- **Next Demonstration:** Live demonstration of the interactive Streamlit batch application.